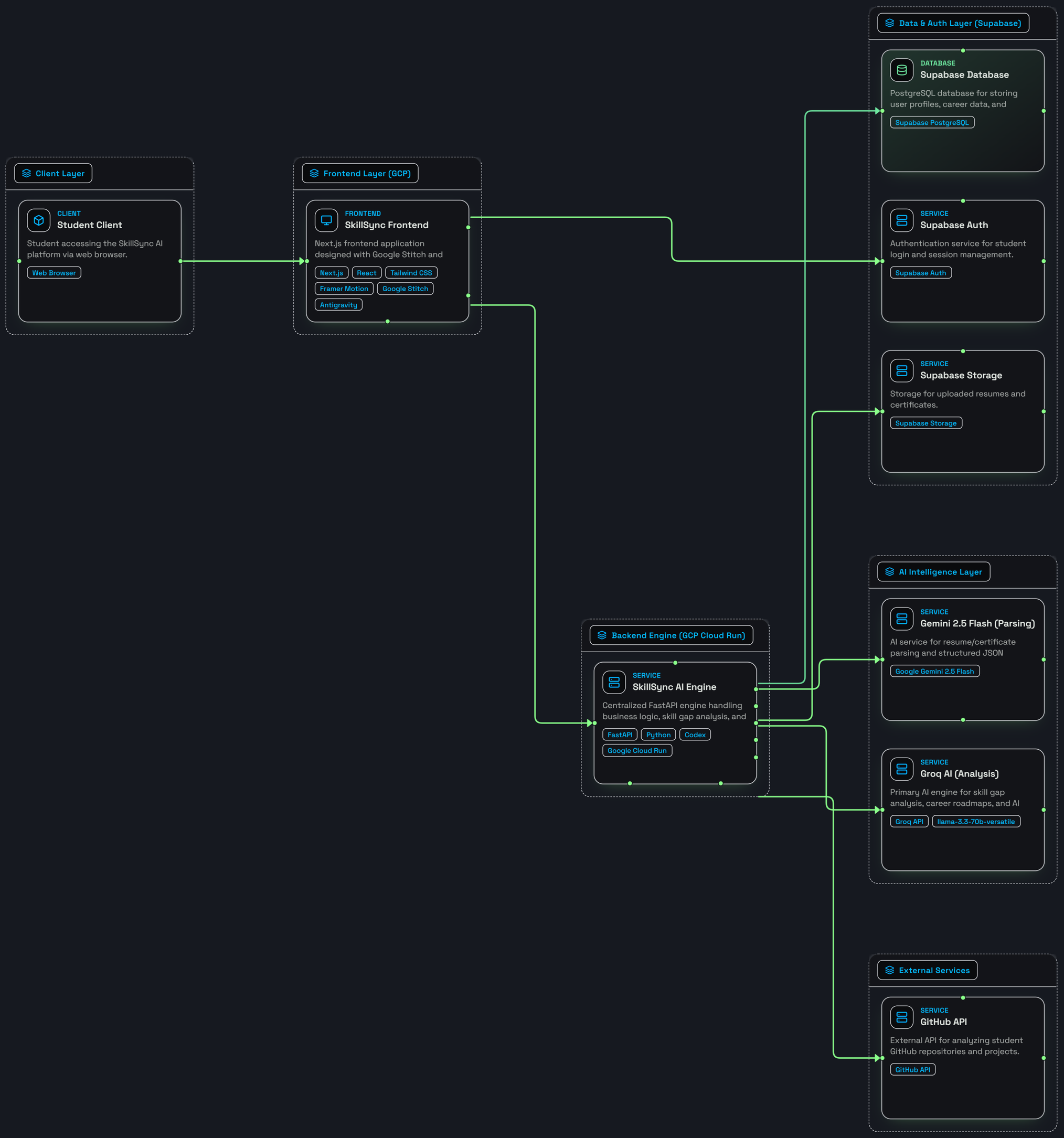
Client Layer
CLIENT
Student Client
Student accessing the SkillSync AI platform via web browser.
Web Browser
Frontend Layer (GCP)
FRONTEND
SkillSync Frontend
Next.js frontend application designed with Google Stitch and
Next.js
React
Tailwind CSS
Framer Motion
Google Stitch
Antigravity
Data & Auth Layer (Supabase)
DATABASE
Supabase Database
PostgreSQL database for storing user profiles, career data, and
Supabase PostgreSQL
SERVICE
Supabase Auth
Authentication service for student login and session management.
Supabase Auth
SERVICE
Supabase Storage
Storage for uploaded resumes and certificates.
Supabase Storage
Backend Engine (GCP Cloud Run)
SERVICE
SkillSync AI Engine
Centralized FastAPI engine handling business logic, skill gap analysis, and
FastAPI
Python
Codex
Google Cloud Run
AI Intelligence Layer
SERVICE
Gemini 2.5 Flash (Parsing)
AI service for resume/certificate parsing and structured JSON
Google Gemini 2.5 Flash
SERVICE
Groq AI (Analysis)
Primary AI engine for skill gap analysis, career roadmaps, and AI
Groq API
llama-3.3-70b-versatile
External Services
SERVICE
GitHub API
External API for analyzing student GitHub repositories and projects.
GitHub API